

QUIZ NO # 2

A cybersecurity team is analyzing the spread of malware across two interconnected systems (System A and System B). The transition of infection between systems after each time cycle is modeled by the matrix:

$$A = \begin{bmatrix} 2 & 1 & 5 & 0 \\ 3 & 9 & 1 & 5 \\ 0 & 5 & 6 & 8 \\ 0 & 0 & 3 & 2 \end{bmatrix}$$

A state vector represents the number of infected files in each system.

A.

Calculate the **eigenvalues** of matrix A and determine the corresponding **eigenvectors**.

B.

Apply the concept of eigenvectors to interpret what it means if an infection state vector remains in the same direction after transformation by A. Explain how this relates to persistent patterns of malware spread between the two systems.

C.

Analyze the eigenvalues and construct a response explaining how they indicate whether the infection is **growing, stabilizing, or decreasing**. Respond with how this analysis can support cybersecurity decision-making in controlling malware propagation.